Title of the Project: AI-Powered Smart Campus Resource Management System

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E1CSA313 Design Project -1

Agenda

1. Introduction

2. Review of Literature

3. Observations from Literature

4. Limitations

5. Problem Statement and Scope

6. Proposed Solution and Objectives

7. Methodology and System Modules

8. Implementation Progress

9. Results and Testing

10. Conclusion and Future Work

11. References

Introduction

Educational campuses need effective resource management to coordinate classrooms, laboratories, utilities and infrastructure. Conventional campus management depends on administrative personnel manually handling timetables and utility schedules, which can make it difficult to prevent clashes and eliminate energy waste promptly.

This project develops a software-based management prototype that visualizes campus spaces in 3D, monitors room occupancy schedules and optimizes electricity and water consumption thresholds. Operating metrics are displayed on a web dashboard to help facility personnel review allocations and support timely decisions.

Review of literature

S.No	Author Name, Journal, Publication Year	Research/ Methodology	Performance Metrics	Advantages	Disadvantages
1	Roda-Sanchez, Cirillo et al., Computer Networks, 2024	CPS with FIWARE broker; ZigBee/LoRaWAN sensor grid in 4 buildings.	34% reduction in data-collection latency	Open-API, interoperable IoT reference architecture.	Static 2D visualization; lacks real-time energy link.
2	Pexyean, Saraubon et al., IEEE Access, 2024	LSTM neural network + discrete-event simulation on smart-meter data.	91.2% load accuracy; 12.5% peak shaving	Validated LSTM load forecasting for campus blocks.	High GPU overhead; no live solar feedback.
3	SIT Research Team, Applied Energy, 2024	Model predictive control + multi-agent dispatch on SCADA/BESS telemetry.	16.2% peak-import reduction; lower degradation	Formal MPC dispatch algorithm for microgrids.	Requires dedicated SCADA; no 3D campus context.

Review of literature

S.No	Author Name, Journal, Publication Year	Research/ Methodology	Performance Metrics	Advantages	Disadvantages
4	Chen, Martinez et al., Automation in Construction, 2024	BIM/IFC spatial data with Revit models + PIR occupancy sensors.	24% booking discrepancy detected	Directly links academic timetables with 3D geometry.	Proprietary BIM format; not web-deployable.
5	Testasecca, Stamatopoulos et al., Energy & Buildings, 2024	EnergyPlus thermodynamic simulation across 48 thermal zones.	14.8% HVAC energy reduction achieved	Calibrated thermodynamic thermal setpoint model.	Single-building only; lacks campus-wide grid view.
6	Saraubon, Pexyean et al., Sensors, 2024	AIoT edge gateways (ESP32/Raspberry Pi) + Modbus telemetry.	94.6% anomaly recall; sub-second alerts	Low-cost open-source edge hardware blueprint.	Offline simulation only; lacks interactive 3D UI.

Review of literature

S.No	Author Name, Journal, Publication Year	Research/ Methodology	Performance Metrics	Advantages	Disadvantages
7	TalTech Energy Group / Campulse, Sust. Cities & Soc., 2024	Power-flow analysis + market bidding across 50+ campus buildings.	19.5% peak shaving via coordinated battery scheduling	Multi-year empirical dataset proving flexibility.	Substation-level only; no occupant spatial link.
8	Scilit / IEEE CSIT Team, IEEE Trans. Smart Systems, 2024	SOA microservices architecture integrating 12 campus services.	28% faster ticket resolution; 8 services unified	Resilient architectural pattern for large universities.	Monolithic backend; latency under concurrent load.
9	Taylor & Francis / Spatial Science, Int. J. Digital Earth, 2024	Drone photogrammetry + LiDAR + WebGL rendering on 40-ha campus.	Millimetre-accurate mesh; interactive browser query	Robust photogrammetry-to-WebGL asset pipeline.	50GB+ dataset; frame drops at high polygon counts.

Review of literature

S.No	Author Name, Journal, Publication Year	Research/ Methodology	Performance Metrics	Advantages	Disadvantages
10	Frontiers / IEEE PES Review, Front. Clean Energy, 2025–2026	PRISMA systematic literature review of 148 papers across 24 nations.	<8% of studies unify 3D spatial view with microgrids	Comprehensive taxonomy of digital-twin frameworks.	Review-based; lacks unified live implementation.

Observations from Literature Review

1. Video and telemetry analysis methods can automate parts of campus resource monitoring and reduce the need for continuous manual observation.

2. Spatial occupancy tracking and schedule analysis are useful for coordinating activities across multi-floor academic areas.

3. Demand monitoring can identify situations where power draw or room vacancy exceeds a predefined operational threshold.

4. Optimization performance depends on data quality, schedule adherence, thermal inertia and campus layout complexity.

LIMITATIONS FROM THE LITERATURE REVIEW

1. Models trained on isolated benchmark datasets may not perform equally well in every real campus environment.

2. Crowded halls, schedule changes and unrecorded cancellations can affect utilization detection.

3. Temporary room absence does not always indicate a vacant class and requires appropriate time context.

4. Continuous 3D spatial rendering and telemetry processing may require considerable computing resources.

Problem Statement / Scope of the Project

PROBLEM STATEMENT

Conventional manual facility tracking places a significant workload on administrative personnel who must coordinate several academic blocks simultaneously. This can delay the identification of vacant classrooms, timetable clashes and other potential utility inefficiencies.

Project Scope

1. Classroom and facility visualization from 3D model input.
2. Monitoring occupancy schedules across predefined academic zones.
3. Energy threshold and vacancy setback alerts.
4. Schedule optimization and dashboard-based review.

Proposed Idea / Solution

The proposed system combines 3D spatial visualization, detection rules and a centralized dashboard to support campus resource management.

Campus data is processed to identify utilization in the monitored area. Configurable rules check vacant-room conditioning and peak energy thresholds. When a rule is triggered, the event is recorded with its timestamp and resource metric. The dashboard presents utilization history for review.

The initial prototype runs on a computer using web technologies, without requiring dedicated IoT hardware.

Objectives of the Idea / Solution as Design Project

1. Develop a software prototype for campus resource management using web input.

2. Monitor classrooms and facilities in the interactive campus view.

3. Automate scheduling to eliminate classroom timetable clashes.

4. Generate alerts when the detected vacancy reaches a configured threshold.

Proposed Research / Methodology

1. Data Input: Ingest timetables and utility records through portal dumps or simulated data.

2. Data Processing: Use Python and Pandas to prepare records for analysis.

3. Space Optimization: Apply constraint satisfaction logic to allocate rooms without clashes.

4. Event Rules: Check room allocations against the schedule boundary and vacancy threshold.

5. Telemetry Logging: Record the timestamp, event type and power metric in a CSV file.

6. Dashboard: Use React and Three.js to display event summaries and utilization history.

7. Testing: Verify allocation generation, logging and dashboard behaviour under different conditions.

HARDWARE REQUIREMENTS

- Laptop or desktop computer for running the application.
- Standard web browser with WebGL support for 3D model rendering.
- Minimum 8 GB RAM recommended for development and testing.
- Adequate storage for software, model files and campus records.

The current prototype uses a computer and browser. Dedicated IoT devices are not required for the implemented modules.

Results and Discussion

The proposed system provides an automated and intelligent approach to campus resource management by analyzing spatial occupancy to identify vacancies, track load curves, and detect unusual or wasteful operations. When an idle conditioned hall is detected, the system generates alerts and stores the telemetry details for facility personnel.

This helps reduce continuous manual monitoring, improve situational awareness, and support a faster response to campus utility anomalies. Further testing can help improve its forecasting and scheduling in real campus environments.

SOFTWARE REQUIREMENTS

1. Python: Application logic and event processing.

2. Three.js: WebGL 3D graphics engine for interactive campus model.

3. React.js: Web frontend for the resource dashboard.

4. Google OR-Tools: Constraint satisfaction timetable optimization.

5. HTML/CSS: Dashboard layout and presentation.

6. CSV: Storage and export of campus utilization records.

7. Visual Studio Code: Application development and code editing.

Proposed Modules

1. 3D Model Input and Room Inspection
Loads campus geometry and visualizes classrooms in the scene.

2. Space Occupancy Monitoring
Checks whether scheduled classrooms are occupied during lecture hours.

3. Energy Threshold Monitoring
Triggers an event when empty rooms remain conditioned past the threshold.

4. Utilization Management
Stores event timestamps, event types and power metrics in a CSV file.

5. Web Dashboard
Displays allocation totals, event categories and telemetry history, with a CSV download option.

IMPLEMENTATION PROGRESS

The initial working prototype has been developed using Python and web technologies.

1. 3D campus visualization runs using a WebGL Three.js model.

2. Timetable scheduling allocations are generated and recorded.

3. Energy alerts are triggered at a configured threshold of fifteen minutes.

4. Telemetry details are stored in `campus_records.csv`.

5. A web dashboard displays resource summaries and schedule history.

6. CSV export functionality has been added.

RESULTS AND TESTING

Initial functional tests confirm that the prototype can process campus data and record timetable allocations and energy threshold events.

1. 3D Visualization: Successfully run using a web browser.

2. Timetable Optimization: Allocation events recorded with timestamps.

3. Energy Management: Alerts generated when the configured threshold is reached.

4. Telemetry Logging: Event information stored in CSV format.

5. Dashboard: Resource summaries and event history displayed.

6. CSV Export: Download option added to the dashboard.

Conclusion & Future Directions

CONCLUSION

A functional software prototype has been developed to support campus resource management and allocation review. The current version integrates browser-based 3D model visualization, zero-clash timetable solving, energy threshold alerts, utilization logging and a web dashboard.

The initial tests demonstrate that these modules can operate together. Further validation is necessary before considering the system ready for deployment in a real campus environment.

FUTURE WORK

1. Improve dashboard visualisations and live meter integration.
2. Test performance under multi-batch schedules and peak load conditions.
3. Evaluate demand prediction reliability and false alerts.
4. Extend the system with additional approved facility modules.
5. Complete documentation, testing and final evaluation.

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